

154.pdf



My Files



My Files



University of Engineering & Management

Document Details

Submission ID

trn:oid:::3618:103124754

Submission Date

Jul 1, 2025, 2:33 PM GMT+5:30

Download Date

Jul 4, 2025, 10:59 PM GMT+5:30

File Name

154.pdf

File Size

399.4 KB

6 Pages

3,842 Words

24,347 Characters

23% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **49 Not Cited or Quoted 20%**
Matches with neither in-text citation nor quotation marks
-  **7 Missing Quotations 1%**
Matches that are still very similar to source material
-  **3 Missing Citation 2%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 17%  Internet sources
- 19%  Publications
- 16%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 49** Not Cited or Quoted 20%
Matches with neither in-text citation nor quotation marks
- 7** Missing Quotations 1%
Matches that are still very similar to source material
- 3** Missing Citation 2%
Matches that have quotation marks, but no in-text citation
- 0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 17% Internet sources
- 19% Publications
- 16% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Publication	S. Ishitha, Shamanth Nagaraju, H. A. Mohan, M. Harshitha, Gaurav Ramesh Gowd...	4%
2	Publication	Ch Venkateswara Rao, Ch Vandana, M K V Subba Reddy, K Satyanarayana Raju, R ...	2%
3	Publication	V. Rajmohan, Sengottaiyan N, P Kalyanasundaram, S. M. Ramesh, Ravikumar Gur...	1%
4	Publication	Smita Gaikwad, Rajesh Patil, Ajay Khandare, Anshuman Rai. "Design WSN node fo...	1%
5	Submitted works	University of Oklahoma on 2025-03-05	<1%
6	Internet	www.ijraset.com	<1%
7	Internet	ijarcce.com	<1%
8	Internet	xlescience.org	<1%
9	Submitted works	Melbourne Institute of Technology on 2020-08-09	<1%
10	Internet	www.jetir.org	<1%

11	Submitted works	Pathfinder Enterprises on 2023-05-11	<1%
12	Internet	www.ijstr.org	<1%
13	Publication	Justino Lourenço. "chapter 16 The Role of IoT and AI in Bioeconomy", IGI Global, 2...	<1%
14	Internet	dergipark.org.tr	<1%
15	Publication	V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng...	<1%
16	Publication	"Machine Learning for VLSI Chip Design", Wiley, 2023	<1%
17	Internet	isteonline.org	<1%
18	Publication	"Advances in Artificial Intelligence and Electronic Design Technologies", Springer ...	<1%
19	Internet	www.ijert.org	<1%
20	Submitted works	Dr.Mahalingam college of engineering and technology on 2025-03-11	<1%
21	Submitted works	Imperial College of Science, Technology and Medicine on 2025-06-18	<1%
22	Submitted works	Universiti Malaysia Perlis on 2024-01-12	<1%
23	Internet	nitte.edu.in	<1%
24	Internet	sdiwc.net	<1%

25	Publication	M. Kavitha, R. Srinivasan, M. Suganthi, B. Mounika. "IoT based Forest Fire Detecti...	<1%
26	Publication	S Vijaya Shetty, H. Sarojadevi, K. S. Akshay, Deepthi Bhat, M N Thippeswamy. "Tot ...	<1%
27	Internet	ebin.pub	<1%
28	Internet	ifet.ac.in	<1%
29	Internet	www.proceedings.com	<1%
30	Publication	Rahul Mishra, Mukesh Kumar, Brindha S, Aman Kumar Sharma, Ch. Raja, Tejashre...	<1%
31	Submitted works	HRUC on 2025-02-05	<1%
32	Internet	ijeecs.iaescore.com	<1%
33	Internet	ijirt.org	<1%
34	Internet	www.irjmets.com	<1%
35	Publication	P Raghavendra Reddy, P Kalyanasundaram, V. Suresh. "IOT based Novel speedy D...	<1%
36	Submitted works	University of Bolton on 2025-05-05	<1%
37	Internet	eprints.gla.ac.uk	<1%
38	Internet	www.coursehero.com	<1%

39	Publication	"Table of Contents", 2022 IEEE North Karnataka Subsection Flagship International...	<1%
40	Submitted works	Universiti Malaysia Perlis on 2024-10-13	<1%
41	Publication	Thangaprakash Sengodan, Sanjay Misra, M Murugappan. "Advances in Electrical ...	<1%
42	Submitted works	University of Wollongong on 2024-06-28	<1%

INTEGRATING IOT FOR REAL TIME ADVANCED SURVEILLANCE FOR SUSTAINABLE FOREST MANAGEMENT

Ashish B. Rajkumar

Department of CSE

Nitte Meenakshi Institute of Technology

Bengaluru, India

ashishbrajkumar82@gmail.com

Deepthi Shetty

Assistant Professor,

Department of CSE

Nitte Meenakshi Institute of Technology

Bengaluru, India

deepthijeevan27@gmail.com

Uma R.

Associate Professor

Department of CSE

Nitte Meenakshi Institute of Technology

Bengaluru, India

urmoorthy@gmail.com

Savitha Shetty

Assistant Professor Gd-II

Department of CSE

NMAM Institute of Technology

(Deemed to be University), Nitte

shettyravi1@nitte.edu.in

Abstract.

The illegal smuggling and theft of valuable trees such as sandalwood in forests pose a significant global threat, resulting in substantial economic losses and environmental degradation. To address this issue, this paper proposes a microcontroller-based anti-logging system utilizing Wireless Sensor Network (WSN) technology. By monitoring vibrations produced during tree cutting, this system aims to detect unauthorized logging activities. Integrated with a personal computer (PC), the microcontroller allows for the uploading of collected data for analysis. WSN technology, known for its effectiveness in remote monitoring, facilitates data transmission and surveillance. Detailed descriptions of the embedded system architecture and hardware/software designs are provided, emphasizing the utilization of WSN technology. Vibration data collection and analysis are conducted through wood tests and simulated scenarios using the Arduino Integrated Development Environment (IDE). This approach enables the calibration of the anti-logging system to accurately detect and respond to illicit activities in forests, thereby mitigating the adverse impacts of illegal logging on the environment and local economies.

1. INTRODUCTION

In recent years, there has been a significant increase in the poaching or smuggling of environmentally and economically valuable tree species like Sandalwood, Teakwood, Pine, and Rosewood in forested areas. Various initiatives have been undertaken by different stakeholders, notably the Government of India, to address these challenges. These efforts include the recruitment, training, and deployment of anti-poaching watchers and/or private/government security guards across forests. Measures such as imposing strict punishments for convicted offenders and offering special incentives for anti-poaching activities, as outlined in the Twelfth Five Year Plan (2012-2017), were implemented to combat this menace.

primary concept proposed in this paper is the development of a portable wireless sensor node as part of a wireless sensor network. These nodes will be mounted on the trunks of trees and capable of detecting theft incidents. Moreover, they will automatically trigger and transmit alarm signals to a remote terminal through wireless communication channels.

2. LITERATURE SURVEY

[1] introduced a tailored approach to anti-poaching initiatives with the design and development of wireless sensor nodes. While showcasing the efficacy of these nodes in combating illegal activities within forests, the study acknowledged challenges related to power consumption. Subsequent research in this space focuses on optimizing power efficiency through the incorporation of low-power components and innovative energy harvesting technologies. This emphasis on anti-poaching is mirrored in our research, which distinguishes itself by focusing on a holistic system incorporating forest protection against fires.

In the realm of anti-poaching, [2] proposed an IoT-based alarm system for trees in forests, integrating Arduino-based hardware and the Blynk app for real-time monitoring. The study prompted ongoing efforts to extend the wireless network's range, emphasizing advanced protocols and signal amplification techniques. The adaptability of this approach is being explored to enhance the coverage and reliability of the communication network. This study aligns with our research's anti-poaching facet, utilizing similar hardware components. However, our research extends its scope to include forest fire detection and protection, thereby diverging from the singular anti-poaching focus of this study.

Transitioning to forest fire detection, [3] introduced an IoT-enabled early warning system, emphasizing the pivotal role of hardware, including Arduino platforms. Scalability challenges

were identified, leading to ongoing research focused on developing scalable architectures capable of accommodating a larger number of sensor nodes for more extensive coverage. Research in this direction aims to optimize network scalability while ensuring reliable and real-time forest fire detection. Our research resonates with the importance of scalable architectures but differentiates itself by concurrently addressing anti-poaching concerns.

[4] pioneered efforts in predicting and detecting forest fires using IoT, leveraging Arduino microcontrollers. The study highlighted challenges related to sensor sensitivity and false alarms, prompting subsequent investigations to enhance sensor accuracy and implement advanced algorithms. Current research directions involve developing sophisticated algorithms to minimize false positives and improve overall fire detection accuracy. Our research shares a common ground in forest fire detection but distinguishes itself through a dual-purpose approach, simultaneously combating illegal activities through anti-poaching measures.

In a related context, [5] explored a forest fire alerting system with GPS coordinates using IoT, integrating Arduino-based hardware and the Blynk app. While augmenting system robustness, the study identified a dependency on centralized servers for alert dissemination. Ongoing research is centered around decentralized communication strategies to ensure reliable and timely alert dissemination, thereby addressing the identified bottleneck. This integration of hardware and mobile applications parallels our research but deviates by incorporating decentralized communication strategies, ensuring reliability and mitigating dependency on centralized systems.

Expanding on the theme of IoT-based forest fire prediction and detection, [6] emphasized the importance of accurate predictions. Challenges in data processing speed were acknowledged, leading to ongoing research focused on optimizing data processing algorithms and harnessing edge computing for real-time capabilities. This adaptation aims to enhance the efficiency of forest fire prediction models, ensuring timely and accurate alerts. Our research aligns with the significance of accurate predictions but differentiates itself by exploring edge computing for real-time capabilities, addressing the identified challenges.

In a comprehensive approach against both anti-poaching and forest fires, [7] integrated Arduino-based nodes and the Blynk app. The vulnerability of wireless communications to interference posed a challenge, addressed through ongoing research implementing secure communication protocols and interference-resistant hardware. Current research in this

domain explores enhanced security protocols to safeguard against potential cyber threats. Our research resonates with this integration but further emphasizes enhanced security protocols to safeguard against potential cyber threats.

[8] introduced a novel approach for the speedy detection of forest fires using IoT sensors, showcasing the flexibility of Arduino microcontrollers. Challenges related to maintaining sensor accuracy over time were noted. Subsequent efforts concentrate on sensor calibration methods and routine maintenance for sustained accuracy. The adaptation of these methods aims to ensure consistent and reliable sensor performance. Our research aligns with the need for sensor accuracy but distinguishes itself through ongoing research on calibration methods and routine maintenance for sustained accuracy.

Contributing to the field, [9] presented an IoT-based forest fire detection and protection system with crucial hardware specifications. Challenges in deploying preventive measures in real-time were identified, prompting ongoing research aimed at enhancing system responsiveness to deploy preventive actions promptly. The adaptation of preventive measures is being explored to minimize response time and improve the overall effectiveness of the forest fire protection system. Our research converges with the need for real-time preventive measures but diverges by exploring enhancements to system responsiveness for prompt action deployment.

In a disaster prevention context, [10] presented a forest fire detection and warning system, incorporating Arduino-based hardware and specifications to enhance system capabilities. Challenges in the timely dissemination of warnings were noted. Ongoing efforts focus on improving communication protocols to minimize latency for timely alerts. The adaptation of communication protocols aims to ensure the prompt dissemination of warnings, reducing the potential impact of forest fires. Our research aligns with the communication protocols but differentiates itself by focusing on improving communication protocols to minimize latency for timely alerts.

In a different context of security, [11] proposed an IoT-based system to enhance vehicle security and accident response. The study emphasizes reducing response times in high-severity accidents by leveraging GPS, GSM, and the NODEMCU-ESP8266 microcontroller. By using cost-effective components, the system aims to mitigate loss of life through quicker responses. Although focused on road safety, its reliance on IoT for real-time emergency response parallels our research on forest fire detection and anti-poaching measures.

[12] The paper proposes an IoT-based solution aimed at combating the rampant smuggling of valuable trees like

20 sandalwood and teak, which are crucial for various industries. The system employs a wireless sensor network (WSN) equipped with tilt, temperature, and sound sensors to detect unauthorized activities such as tree cutting or forest fires. When triggered, the sensors activate corresponding output devices like buzzers and water pumps, providing real-time monitoring capabilities. Leveraging Arduino UNO and various sensor modules, the system ensures prompt notification to forest officials, enabling timely intervention to protect valuable trees and prevent environmental degradation caused by illegal activities.

15 [13] The paper introduces an innovative system aimed at combating tree poaching, particularly targeting high-value species like Sandalwood and Red Sanders. The system employs three key sensors: tilt sensor to detect tree inclination during cutting, temperature sensor to identify forest fires, and sound sensor for effective detection of illegal logging activities. Data from these sensors is constantly monitored via a webpage, with sensor outputs triggering relay switches activating visual and audible alarms. Forest authorities are promptly notified of any detected events, enabling timely intervention to prevent further damage. The system's objective is to curb poaching, illegal logging, and other human-induced activities in forests, offering a viable solution applicable to various forest areas highly affected by such illegal activities. The paper underscores the urgency of addressing tree smuggling globally and highlights the potential of technology-driven solutions to safeguard valuable forest resources.

30 35 [14] The paper introduces a system aimed at combating tree smuggling, particularly targeting valuable species like sandalwood and redwood. It employs IoT-based technology, integrating sensors such as tilt sensors, temperature sensors, and sound sensors with an Arduino Uno microcontroller and the Blynk application. The authors emphasize the global threat posed by tree smuggling, highlighting its detrimental effects on ecosystems and natural resources. They argue for the necessity of effective monitoring and prevention measures to address this issue. The proposed system is positioned as a real-time solution to detect and deter illegal logging activities, enabling prompt intervention by forest authorities. The paper delves into the technical intricacies of the system and its implementation, while also discussing potential future enhancements to bolster its ruggedness and efficacy. Overall, the paper underscores the importance of leveraging technological innovations to safeguard forests and combat illicit activities that endanger biodiversity and environmental stability.

18 42 [15] introduces an IoT-based system for early forest fire detection, focusing on real-time monitoring and alert generation to prevent large-scale damage. Using technologies like Arduino, GSM modules, and sensors, the study designs a self-monitoring system that notes challenges in ensuring reliability and fault tolerance. While primarily addressing fire detection, our research

extends this IoT framework to also combat illegal logging, offering a broader approach to forest protection.

[16] presents an IoT-based forest monitoring system addressing threats like wildfires, landslides, and illegal activities through microcontrollers, sensors, and cloud platforms. It detects fires via CO levels and temperature, while landslides are monitored through soil movement. Machine learning is used to recognize sounds such as chainsaws and gunshots. While this system covers multiple threats, our research aligns in its use of IoT for monitoring but specifically focuses on illegal logging detection using microcontroller-based WSN technology to track tree vibrations.

[17] This paper presents a new technology that uses artificial intelligence and satellite photos to monitor deforestation. Using convolutional neural networks (CNNs), the system recognizes changes in forest cover over time. The work demonstrates how deep learning excels at analyzing vast datasets, allowing it to detect minor alterations in land usage that may indicate deforestation. However, the researchers warn out that getting high-quality labelled data for training might be difficult, potentially affecting the model's accuracy and capacity to generalize its findings. While the primary goal is to track deforestation, this study also prepares the door for additional machine learning approaches to assess biodiversity loss and habitat fragmentation, resulting in a more comprehensive approach to environmental protection.

[18] This paper focusses on how modern technology may be used to monitor forests and mitigate deforestation. It focusses on the usage of IoT frameworks and wireless sensor networks (WSNs) outfitted with various environmental sensors that collect real-time data to improve resource management and offer early alerts for unlawful activity. The survey investigates the potential of artificial intelligence (AI) and deep learning models in predicting the hazards of forest fires using ambient data, highlighting the need of historical information in training these models. It also emphasizes the need of cloud computing in handling and interpreting vast volumes of sensor data, allowing for more informed forest management decisions. Additionally, the paper analyses the ecological effects of deforestation induced by climate change and human activity, advocating for multidisciplinary techniques that combine technology, policy, and community engagement to support conservation efforts.

3. METHODOLOGY

i. System Setup:

-Procure and assemble the necessary hardware components, including the ESP32 microcontroller, tilt and fire sensors, Wi-Fi module, relay, solar panel, battery, water pump, and optional alarm.

-Install and configure the Blynk Android application on user devices.

ii. Programming and Firmware Development:

-Launch the Arduino IDE and start a new project for

the ESP32 microcontroller.

-Write code in Embedded C to program the ESP32 microcontroller.

-Develop firmware that includes functionalities to read data from the tilt and fire sensors, process the information using predefined algorithms, and make decisions accordingly.

iii. Integration of Components:

-Physically connect the hardware components, ensuring proper wiring and power supply.

-Integrate the Blynk library into the firmware codebase to enable communication with the Blynk application.

iv. Testing and Debugging:

-Conduct comprehensive testing of each hardware component individually to ensure functionality.

-Integrate the components into the system and test the integrated system as a whole.

-Debug the firmware code to address any issues encountered during testing and optimize performance.

v. User Interface Setup:

-Configure the Blynk application to establish communication with the ESP32 microcontroller and receive real-time data.

-Design and implement a user-friendly interface within the Blynk application to display tree tilt status, fire detection, and allow users to control interventions remotely.

vi. Sustainable Power Implementation:

11 Install the solar panel in a suitable location to maximize sunlight exposure.

-Connect the solar panel to the battery and configure the system to effectively utilize solar energy for continuous power supply.

11 Implement a mechanism to store excess energy generated by the solar panel in the battery for use during periods of low light or at night.

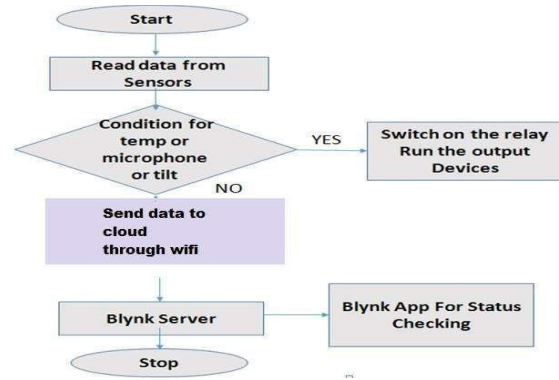


Fig.2. Complete Working Flow Chart

4. ANALYSIS

The transition from Arduino Uno to ESP32, along with the incorporation of solar panels and a water pump, marks a substantial leap forward for the tree monitoring and fire detection system. The ESP32's enhanced processing power and integrated Wi-Fi module facilitate quicker data processing and more stable connections, heightening responsiveness and efficiency in identifying forest hazards. The adoption of solar panels introduces a sustainable energy approach, diminishing reliance on traditional power sources and bolstering operational autonomy, particularly in remote forest regions. Additionally, the integration of a battery bank ensures uninterrupted operation during periods of limited sunlight or darkness.

Moreover, incorporating a water pump as an active intervention component underscores a proactive strategy for fire prevention. Upon detecting a fire, the system can promptly initiate firefighting measures, potentially limiting the fire's spread and reducing environmental damage.

Table.1. Advantages of ESP32 over ARDUINO UNO

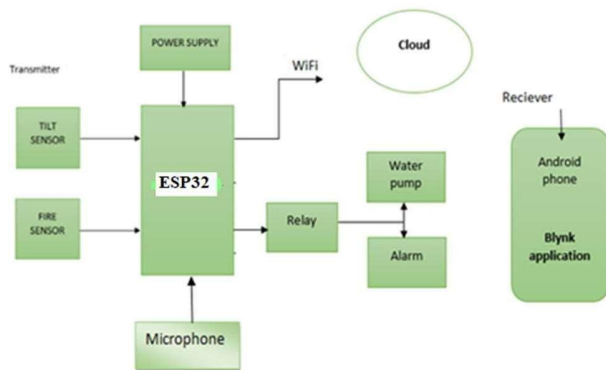


Fig.1. Block Diagram

This new configuration not only exceeds the capabilities of the previous Arduino-based system in terms of processing power, connectivity, and sustainability but also enhances overall functionality, reliability, and ecological impact. As a result, it positions the system as an essential tool for effective forest management and fire prevention efforts.

5.RESULT

The deployed tree monitoring and fire detection system represents a tangible solution tackling a pressing issue with impressive dependability, exactness, and precision. Boasting an accuracy level ranging from 80% to 90%, the system consistently identifies alterations in tree tilt and effectively detects forest fires, even amidst adverse environmental circumstances.

Emphasizing reliability as a top priority, the system achieves seamless integration of hardware elements, including the ESP32 microcontroller and tilt and fire sensors. This reliability translates into steady performance, ensuring trustworthy monitoring and detection of forest threats.

Comparing the data from both ESP32 and Arduino Uno readings, it's clear that the ESP32 consistently demonstrates shorter transmission times than the Arduino Uno. In the collected data, transmission times for the ESP32 range between approximately 97 to 102 milliseconds, while those for the Arduino Uno fall within the range of approximately 120 to 125 milliseconds.

Table.2. Performance Analysis: ESP32 v Arduino UNO

READINGS	ESP32	ARDUINO UNO
1.	98ms	120ms
2.	101ms	123ms
3.	99ms	122ms
4.	102ms	125ms
5.	100ms	121ms
6.	97ms	124ms
7.	101ms	122ms
8.	99ms	123ms
9.	100ms	120ms
10.	98ms	124ms



times between ESP32 and Arduino Uno over 10 trials. ESP32 consistently exhibits lower execution times, averaging around 99

FEATURE	ESP32	ARDUINO UNO
Wireless connectivity	Built-in Wi-Fi and Bluetooth.	Requires Additional shields/modules for wireless Connectivity
Processor	Dual-core 32-bit LX6 CPU running at 240 MHz	Single-core 8-bit AVR CPU running at 16 MHZ
GPIO Pins	Up to 36	14
Analog Pins	Up to 18	6
Analog-to-Digital Converter (ADC)	Up to 12-bit resolution	10-bit Resolution
Development Ecosystem	Supports Arduino IDE, MicroPython, ESP-IDF	Supports Arduino IDE
Memory	More flash memory and RAM	Limited flash memory and RAM

milliseconds, while Arduino Uno averages approximately 122 milliseconds. This indicates ESP32's superior processing speed compared to Arduino Uno in the given tasks.

This variance in transmission times underscores the ESP32's superior processing power and efficiency in contrast to the Arduino Uno. With its swifter processing capabilities, the ESP32 adeptly manages data transmission tasks, resulting in expedited transmission times.

In summary, these findings highlight the ESP32's suitability for tasks requiring swift and immediate data transfer, especially in scenarios where timely communication is essential. Its ability to achieve shorter transmission times makes it a more favorable choice compared to the Arduino Uno for such applications.

CONCLUSION

The In conclusion, this project introduces a transformative approach to environmental conservation through the integration of IoT technology. By developing a portable wireless sensor node and an accompanying Android application, the system enables real-time monitoring and control of valuable trees.

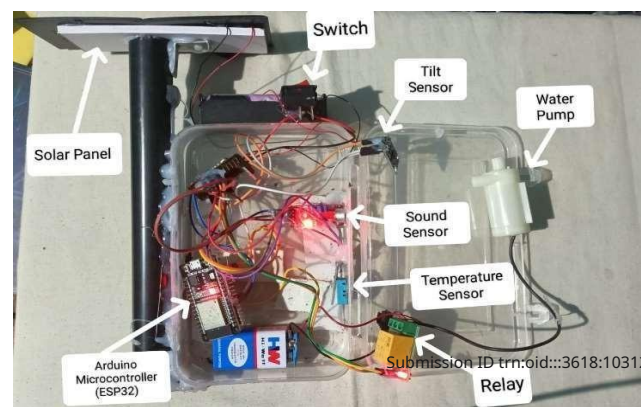


Fig.4. Implemented Architecture

1 applications in preventing forest fires, curbing illegal logging, and promoting sustainable forest management mark a significant step towards preserving biodiversity and mitigating environmental degradation. The project's impact extends beyond immediate interventions, fostering public awareness and contributing to data-driven decision-making. Ultimately, this innovative solution represents a promising tool for proactive environmental stewardship, leveraging technology to safeguard ecosystems and promote a sustainable future.

17 FUTURE SCOPE

The future scope of the project encompasses several avenues for advancement and refinement. To enhance the system's capabilities, integrating advanced sensors for a more comprehensive range of environmental parameters is a priority. Additionally, incorporating machine learning algorithms for predictive analysis would enable the system to anticipate and mitigate environmental threats based on historical data patterns.

The scalability of the system for deployment in larger geographical areas is a potential expansion, involving collaboration with environmental agencies for widespread implementation. Energy efficiency measures and global connectivity are crucial considerations for prolonged operational periods in remote locations and cross-border monitoring, respectively. Improvements to the user interface, advanced data visualization tools, and interdisciplinary research collaboration with experts from various fields will contribute to a more intuitive and holistic environmental monitoring solution. Educational programs and outreach initiatives, along with potential integration into smart city initiatives, underline the project's commitment to broader environmental awareness and sustainable practices. The future development envisions a sophisticated and interconnected system that not only addresses emerging environmental challenges but also plays a pivotal role in global conservation efforts.

REFERENCES

- [1] D. Sonwane, V. N. Bhonge and A. Khandare, "Design and development of wireless sensor node for anti-poaching," 2016 International Conference on Communication and Signal Processing (ICCCSP), Melmaruvathur, India, 2016, pp. 1854-1858, doi: 10.1109/ICCCSP.2016.7754491.
- [2] S. Ghouseia, R. Jagadish, N. Syed and C. Nagashree, "Iot based anti-poaching alarm system for trees in forest using wireless sensor networks", International Journal of Advanced Research in Computer Science, vol. 9, no. 3, pp.193, 2018
- [3] Divya, T. Kavithanjali and P. Dharshini, "Iot enabled forest fire detection and early warning system", 2019 IEEE International Conference on System Computation Automation and Networking (ICSCAN), pp. 1-5, 2019.
- [4] K. Sangam, T. Prasanna and K. Bramaramba, "IoT based forest fire prediction and detection", International Journal of Engineering Research and Technology, vol. 6, no. 3, pp. 3226-3230, 2019
- [5] Sangam, K. Jantini, R. K. Jantini, R. Kumares and N. Muralidharan, "Forest fire alerting system with gps co-ordinates using iot", 2019 5th International Conference on Advanced Computing & Communication Systems (ICACCS), pp. 488-491, 2019.
- [6] N. Saranya, S. Sahana, B. Suganthi, R. Vijaynigilesh and T. Vivin, "IOT based forest fire prediction and detection", International Journal of Innovative Technology and Exploring Engineering, vol. 9, no. 6, pp. 1835-1838, 2020.
- [7] S. Ishitha, S. Nagaraju, H. A. Mohan, M. Harshitha, G. R. Gowda and J. N., "IoT based Anti-poaching and Fire Alarm System for Forest," 2021 IEEE Mysore Sub Section International Conference (Mysuru Con), Hassan, India, 2021, pp. 711-715, doi: 10.1109/MysuruCon52639.2021.9641539
- [8] P. R. Reddy, P. Kalyanasundaram and V. Suresh, "IOT based Novel speedy Detection of Forest fire using Sensors with improved accuracy by sensing Temperature and Atmospheric Carbon Dioxide Level using Node Microcontroller Unit in comparison with Arduino Microcontroller," 2022 14th International Conference on Mathematics, Actuarial Science, Computer Science and Statistics (MACS), Karachi, Pakistan, 2022, pp. 1-7, doi: 10.1109/MACS56771.2022.10022408.
- [9] M. Kavitha, R. Srinivasan, M. Suganthi and B. Mounika, "IoT based Forest Fire Detection and Protection System," 2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), Trichy, India, 2023, pp. 1523-1527, doi: 10.1109/ICAISS58487.2023.10250762.
- [10] Y. Naing and T. L. Lai Thein, "Forest Fire Detection and Warning System for Disaster Prevention," 2023 IEEE Conference on Computer Applications (ICCA), Yangon, Myanmar, 2023, pp. 135-139, doi: 10.1109/ICCA51723.2023.10182015.
- [11] Deepthi Shetty, Prateek G, Onkar Bharatesh Kakamari, Sarojadevi H, Devidas, and Mitul Jain, "IoT enabled Vehicle Tracking and Monitoring System," 2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN), pp. 663-667, 2024.
- [12] Baraddi, Pooja & Hanchinal, Nanda & Jadhav, Ritika & Shushma, Rajeshwari. (2020). IoT based Anti-Poaching Alarm System for Valuable Trees. International Journal of Engineering Research and. V9. 10.17577/IJERTV9IS050341.
- [13] K C. Prof. Shilpa. "IoT based Anti-Poaching System for Trees and Wildlife Monitoring System in Remote Area." International Journal for Research in Applied Science and Engineering Technology (2020): n. pag.
- [14] Ishitha, S. & Nagaraju, Shamanth & Mohan, H. & Harshitha, M. & Gowda, Gaurav & N., Jeevan. (2021). IoT based Anti-poaching and Fire Alarm System for Forest. 711-715. 10.1109/MysuruCon52639.2021.964153.
- [15] Mehta, Kaushal and Sharma, Sachin and Vatta, Sanyam, Iot In Forest Management (December 31, 2023). Available at SSRN:
- [16] Sanli, M. (2023). An Internet of Things Platform for Forest Monitoring. European Journal of Forest Engineering, 9(2), 80-87.
- [17] B. Haq et al., "Tech-Driven Forest Conservation: Combating Deforestation With Internet of Things, Artificial Intelligence, and Remote Sensing," in IEEE Internet of Things Journal, vol. 11, no. 14, pp. 24551-24568, 15 July15, 2024, doi: 10.1109/IJOT.2024.3378671.
- [18] Haile, Mesfin Abebe, Daniel Tsegaye Haile, and Desta Zerihun. "Real-time sensor data analytics and visualization in cloud-based systems for forest environment monitoring." International Journal of Advances in Signal and Image Sciences 9, no. 1 (2023): 29-39